

Vaishnav Mukund Mankar

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SUMMARY

Computer Engineering undergraduate focused on building high-performance backend systems and reliable IoT data pipelines. Experienced in bridging physical hardware with cloud infrastructure, optimizing software for speed, and utilizing modern development workflows to deliver scalable production solutions.

TECHNICAL SKILLS

Languages: Python, SQL, Rust, Go, TypeScript

Cloud & Tools: AWS, Docker, Git, Linux, TimescaleDB, PostgreSQL, Redis, Firebase, OpenTelemetry (OTLP)

Concepts: Data Structures & Algorithms (DSA), Operating Systems (OS), Database Management Systems (DBMS), Computer Networks (CN), Object-Oriented Programming (OOP), System Design

EXPERIENCE

Backend & Systems Engineer Intern

May 2026 – Present

Golain

Remote/Hybrid

- Engineered a high-performance EV proxy system in Rust for **Enough Energy (UK)** to connect physical charging stations with central software utilizing the OCPP-J 1.6 protocol over WebSockets.
- Designed a low-latency messaging loop using ZeroMQ (ZMQ) and MQTT to route real-time telemetry, using an embedded Sled DB cache to prevent telemetry data loss during temporary network drops.
- Leveraged AI-agent workflows to generate boilerplate and test suites under direct system design control, accelerating deployment speed while owning database schemas, query optimizations, and container orchestration.

Freelance Full-Stack Developer & Agent Operator

April 2026

Independent

Pune, India

- Designed dynamic KPI charts, visualizations, and client-side AI summarizers to provide real-time strengths, weaknesses, and writing dynamics analysis.
- Managed AI coding agents to rapidly scaffold UI components and routing, focusing manual cognitive effort on system schema, database design, and hardware-software data mapping.

AI Developer Intern

Aug 2025 – May 2026

Banao.tech

Remote

- Engineered backend components for an AI-driven EdTech video generation pipeline, automating personalized multilingual video creation from images and textual summaries.
- Learned and applied python-langchain, GitHub workflows, AWS Lambda serverless execution, and cloud logging telemetry to optimize execution and debug latency.

PROJECTS

FieldPulse (Industrial IoT Backend Platform) Tech Stack: Go, gRPC, TimescaleDB, Redis, MQTT, OpenTelemetry (OTLP)

- Developed a distributed industrial IoT telemetry ingestion backend in Go, utilizing gRPC and TimescaleDB.
- Integrated Redis for sensor telemetry deduplication and alert suppression, and instrumented all services with OpenTelemetry to trace across HTTP, gRPC, and MQTT boundaries.

AI-Powered Smart Photo Gallery (digital-asset-management) Tech Stack: React, FastAPI, PostgreSQL, Gemini, Docker

- Engineered a private, local-first "Google Photos" alternative allowing natural language search queries on unstructured media, implementing AI-driven semantic image tagging and text-to-SQL search query optimizations.

ACHIEVEMENTS

IEEE Publication: Published "GNN-Based Friend Recommendation: A Graph Embedding Approach for Social Networks" in an IEEE International Conference.

EDUCATION

Pune Institute of Computer Technology (PICT)

Pune, India

B.E. Computer Engineering; CGPA: 9.2

2023 – 2027

Global Public School

Pune, India

Class XII (HSC); Score: 80%

2023

Ryan International School

Pune, India

Class X (SSC); Score: 90%

2021